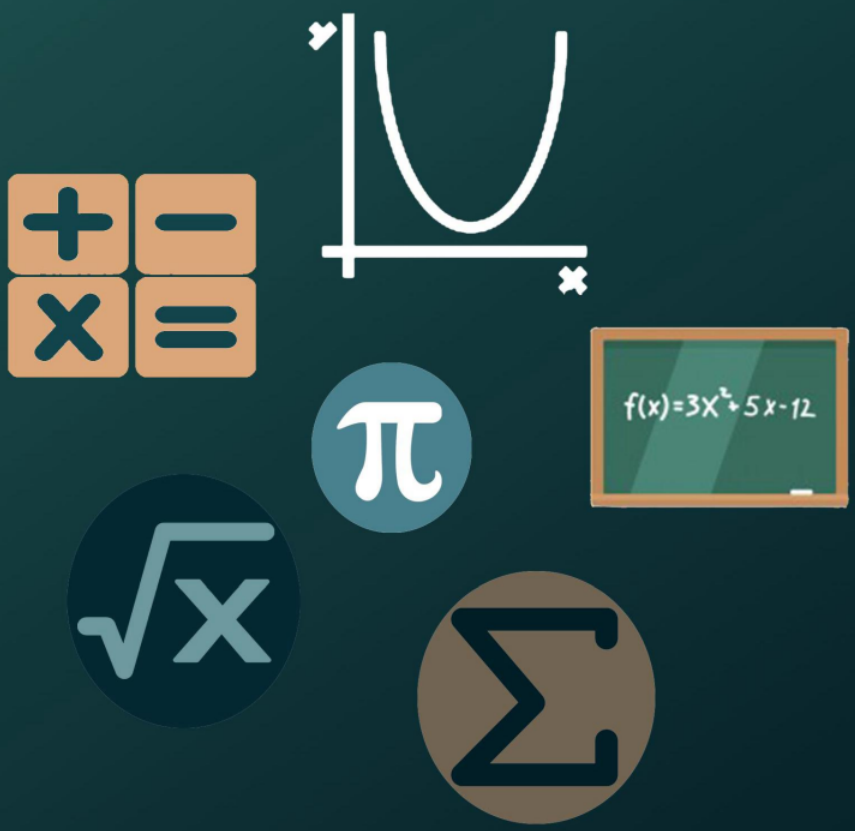
MATHEMATICS

Methods of Differentiation



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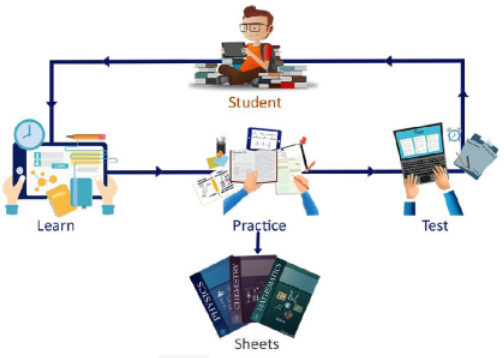
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Derivative by First Principle		15		23
2	Derivative of Standard Functions	1-4	1	1	1
3	Theorems of Differentiation	5-10	2-3		19-20
4	Chain Rule	11-18	4,17	2	2
5	Logarithmic Differentiation	19-20	5,23-26	3	
6	Parametric Differentiation	21-25	10-11	4-5	3,21
7	Derivative of One Function w.r.t. Another Function	26-29	6	6-8	
8	Derivative of Implicit Function	30-32	18	9	4-5,24
9	Derivative of Infinite Series	33	7,19-20	10-11	
10	Derivative of Homogeneous Equation	34			
11	Derivative of Inverse Functions	35	27	12	6,22
12	Derivatives of Inverse Trigonometric Functions	36-40	8,15	13-17	17,25
13	Successive Differentiation	41-46	28-29	18-23	7,11-13, 26
14	L'Hospital Rule	47-52	16,21	24	14-16
15	Derivative of Functions Expressed in the Determinant Form	53-54		25	8
16	Deduction of New Identities by Differentiating a Given Identity				9-10,27
17	After Completing Chapter		9,12-14, 22		18,28

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Derivative of Standard Functions

Q1. If $f(x)=\left(\frac{x^a}{x^b}\right)^{a+b}\cdot\left(\frac{x^b}{x^c}\right)^{b+c}\cdot\left(\frac{x^c}{x^a}\right)^{c+a}$, then $f'(x)$ is equal to -

(A) 1 (B) 0

(C) x^{a+b+c} (D) None of these

Q2. If $f(x)=\ell n|2x|$, $x\neq 0$, then $f'(x)$ is equal to -

(A) $\frac{1}{x}$ (B) $-\frac{1}{x}$

(C) $\frac{1}{|x|}$ (D) None of these

Q3. If $y=(1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$, then $\frac{dy}{dx}$ is equal to -

(A) 1 (B) –1 (C) x (D) $\sqrt{x}$

Q4. The derivative of $\tan x^\circ$ is -

(A) $\left(\frac{\pi}{180}\right)\cos^2 x$ (B) $\left(\frac{\pi}{180}\right)\sec^2 x$

(C) $\left(\frac{\pi}{180}\right)\cos^2 x^\circ$ (D) $\left(\frac{\pi}{180}\right)\sec^2 x^\circ$

Theorems of Differentiation

Q5. If $f(x)=|\cos x|$, then $f'\left(\frac{3\pi}{4}\right)$ is equal to -

(A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$

(C) 1 (D) None of these

Q6. If $\frac{d}{dx}\left(\frac{1+x^4+x^8}{1+x^2+x^4}\right)=ax^3+bx$ then -

(A) a = 4, b = 2 (B) a = 4, b = –2

(C) a = –2, b = 4 (D) None of these

Q7. If $f(x)=(ax+b)\sin x+(cx+d)\cos x$, then the values of a, b, c and d such that $f'(x)=x\cos x$ for all x are -

(A) b = c = 0, a = d = 1

(B) b = d = 0, a = c = 1

(C) c = d = 0, a = b = 1

(D) None of these

Q8. If $f(x)=|\cos x-\sin x|$, then $f'\left(\frac{\pi}{2}\right)$ is equal to -

(A) 1 (B) –1

(C) 0 (D) None of these

Q9. If $f(x)=\frac{1-\cos x}{1-\sin x}$, then $f'\left(\frac{\pi}{2}\right)$ is equal to-

(A) 1 (B) 0

(C) ∞ (D) does not exist

Q10. The differential coefficient of $\frac{x\sin x}{1+\cos x}$ is-

(A) $\frac{x-\sin x}{1+\cos x}$ (B) $\frac{x+\sin x}{1+\cos x}$

(C) $\frac{x-\sin x}{(1+\cos x)^2}$ (D) $\frac{x+\sin x}{(1+\cos x)^2}$

Chain Rule

Q11. If $y=e^{\tan x}$, then $\cos^2 x\,\frac{d^2y}{dx^2}=$

(A) $(1-\sin 2x)\frac{dy}{dx}$ (B) $-(1+\sin 2x)\frac{dy}{dx}$

(C) $(1+\sin 2x)\frac{dy}{dx}$ (D) None of these

Q12. If $f(x) = \log_x(\ell n\, x)$, then $f'(x)$ at $x = e$ is

(A) e (B) $\frac{1}{e}$

(C) $\frac{2}{e}$ (D) 0

Q13. If $y = \log\left(\frac{e^x}{e^x+1}\right)$, then dy/dx equals-

(A) $\frac{1}{e^x+1}$ (B) $\frac{1}{(e^x+1)^2}$

(C) $\frac{e^x-1}{e^x+1}$ (D) None of these

Q14. If $y = \frac{\sec x - \tan x}{\sec x + \tan x}$, then $\frac{dy}{dx}$ equals-

(A) $2\sec x\,(\sec x - \tan x)^2$

(B) $-2\sec x\,(\sec x - \tan x)^2$

(C) $2\sec x\,(\sec x + \tan x)^2$

(D) $-2\sec x\,(\sec x + \tan x)^2$

Q15. If $y = \log_x 10$, then the values of dy/dx equals-

(A) 1/x (B) 10/x

(C) $-\frac{(\log_x 10)^2}{x \log_e 10}$ (D) $\frac{1}{(x \log_e 10)}$

Q16. The differential coefficient of $\log\tan(x/2 + \pi/4)$ is-

(A) cosecx (B) cosx

(C) secx (D) sinx

Q17. If $y = \log(\sec x + \tan x)$ then $\frac{dy}{dx}$

(A) tanx (B) secx

(C) $\tan^2 x$ (D) None of these

Q18. If $\phi(x) = \log_5 \log_3 x$, then $\phi'(e)$ is equal to-

(A) elog5 (B) – elog5

(C) $\frac{1}{e \log 5}$ (D) None of these

Logarithmic Differentiation

Q19. Find $\frac{dy}{dx}$ at $x = 0$, when

$y = (1+x)(1+x^2)(1+x^4)\ldots(1+x^{2^{n-1}})\,(1+x^{2n})$,

(A) –1 (B) 1

(C) 0 (D) None of these

Q20. If $y = (\sin x)^{\cos^{-1} x}$, then $\frac{dy}{dx}$ =

(A) $(\sin x)^{\cos^{-1} x}\left(\cos^{-1} x.\cot x - \frac{\log \sin x}{\sqrt{1-x^2}}\right)$

(B) $(\sin x)^{\cos^{-1} x}\left(\cos^{-1} x.\cot x + \frac{\log \sin x}{\sqrt{1-x^2}}\right)$

(C) $(\sin x)^{\cos^{-1} x}\left(\cos^{-1} x.\tan x + \frac{\log \sin x}{\sqrt{1-x^2}}\right)$

(D) None of these

Parametric Differentiation

Q21. If $x = f(t)$, $y = \phi(t)$, then $\frac{d^2y}{dx^2}$ is equal to

(A) $\frac{f_1\phi_2 - \phi_1 f_2}{f_1^2}$ (B) $\frac{f_1\phi_2 - \phi_1 f_2}{f_1^3}$

(C) $\frac{\phi_1 f_2 - f_1\phi_2}{f_1^3}$ (D) None of these

Q22. If $x = a(\cos\theta + \theta \sin\theta)$,

$y = a(\sin\theta - \theta\cos\theta)$, $\frac{dy}{dx}$ =

(A) $\cos\theta$ (B) $\tan\theta$

(C) $\sec\theta$ (D) $\operatorname{cosec}\theta$

Q23. If $x = a\sin t$, $y = a\cos t$ then $\frac{d^2y}{dx^2}$ equals-

(A) $-1/a\cos^3 t$ (B) $-1/a\sec^3 t$

(C) $-\cos^3 t$ (D) None of these

Q24. If $x = at^2$ and $y = 2at$ then $\frac{dy}{dx} =$

(A) t (B) $\frac{1}{t}$

(C) t^2 (D) $-1/t^2$

Q25. If $x = 3\cos\theta - 2\cos^3\theta$ and $y = 3\sin\theta - 2\sin^3\theta$, then $\frac{dy}{dx} =$

(A) $\sin\theta$ (B) $\cos\theta$

(C) $\tan\theta$ (D) $\cot\theta$

Derivative of One Function w.r.t. Another Function

Q26. The derivative of f(tan x) w.r.t. g(sec x) at $x = \frac{\pi}{4}$, where $f'(1) = 2$ and $g'(\sqrt{2}) = 4$, is -

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$

(C) 1 (D) None of these

Q27. The derivative of $\sin^{-1}x$ with respect to x^2 is-

(A) $\frac{1}{x\sqrt{1-x^2}}$ (B) $\frac{1}{2\sqrt{1-x^2}}$

(C) $\frac{1}{2x\sqrt{1-x^2}}$ (D) $-\frac{1}{\sqrt{1-x^2}}$

Q28. The 2nd derivative of $a\cos^3t$ w.r.t. $a\sin^3t$ at $t = \frac{\pi}{4}$ is-

(A) $\frac{4\sqrt{2}}{3a}$ (B) 2

(C) $\frac{1}{12a}$ (D) 0

Q29. The differential co-efficient of e^{x^3} w.r.t. logx is-

(A) $3x^2e^{x^3} + 3x^2$ (B) e^{x^3}

(C) $3x^3e^{x^3}$ (D) $3x^2e^{x^3}$

Derivative of Implicit Function

Q30. If $e^x + e^y = e^{x+y}$, then the values of $\frac{dy}{dx}$ at (1, 1) is -

(A) 0 (B) –1

(C) 1 (D) None of these

Q31. If $x^y = e^{x-y}$, then $\frac{dy}{dx}$ is equal to-

(A) $\frac{\log x}{(1+\log x)^2}$ (B) $\frac{x-y}{(1+\log x)}$

(C) $\frac{x-y}{(1+\log x)^2}$ (D) $\frac{1}{1+\log x}$

Q32. If $y = \log_y x$, then $\frac{dy}{dx} =$

(A) $\frac{1}{x+\log y}$ (B) $\frac{1}{\log x(1+y)}$

(C) $\frac{1}{x(1+\log y)}$ (D) $\frac{1}{y+\log x}$

Derivative of Infinite Series

Q33. If $x = e^{y+e^{y+e^{y+\ldots\infty}}}$, $x > 0$, then $\frac{dy}{dx}$ is -

(A) $\frac{x}{1+x}$ (B) $\frac{1}{x}$

(C) $\frac{1-x}{x}$ (D) $\frac{1+x}{x}$

Derivative of Homogeneous Equation

Q34. If $x^m.y^n = (x + y)^{m+n}$, then $\frac{dy}{dx}$ is-

(A) $\frac{x+y}{xy}$ (B) xy

(C) $\frac{x}{y}$ (D) $\frac{y}{x}$

Derivative of Inverse Functions

Q35. If g is the inverse of f and $f'(x) = \frac{1}{1+x^3}$, then $g'(x)$ is equal to

(A) $1 + [g(x)]^3$ (B) $\frac{1}{1+[g(x)]^3}$

(C) $[g(x)]^3$ (D) $1 + x^3$

Derivatives of Inverse Trigonometric Functions

Q36. If $y = \tan^{-1}\frac{x-\sqrt{1-x^2}}{x+\sqrt{1-x^2}}$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{-1}{\sqrt{1-x^2}}$ (B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{-x}{\sqrt{1-x^2}}$ (D) None of these

Q37. The differential coefficient of $\tan^{-1}\frac{2x\sqrt{1-x^2}}{1-2x^2}$ w.r.t. $\sec^{-1}\frac{1}{2x^2-1}$ at $x = \frac{1}{2}$ is equal to -

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$

(C) -1 (D) None of these

Q38. $\frac{d}{dx}\left(\sin^2\left(\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)\right)$ equals-

(A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 1

Q39. If $y = \sec^{-1}\frac{\sqrt{x}-1}{x+\sqrt{x}} + \sin^{-1}\frac{x+\sqrt{x}}{\sqrt{x}-1}$, then $\frac{dy}{dx}$ is equal to-

(A) x (B) 1

(C) 0 (D) None of these

Q40. If $y = \cos^{-1}\left(\frac{3\cos x - 2\sin x}{\sqrt{13}}\right)$, then $\frac{dy}{dx}$ =

(A) 1 (B) 0

(C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{13}}$

Successive Differentiation

Q41. Let f(x) be a polynomial of degree 3 such that $f(3) = 1$, $f'(3) = -1$, $f''(3) = 0$ and $f'''(3) = 12$. Then the value of $f'(1)$ is

(A) 12 (B) 23

(C) –13 (D) None of these

Q42. If $y = (\sin^{-1}x)^2$, then $(1 - x^2)\frac{d^2y}{dx^2}$ is equal to

(A) $x\frac{dy}{dx} + 2$ (B) $x\frac{dy}{dx} - 2$

(C) $-x\frac{dy}{dx} + 2$ (D) None of these

Q43. If $y^2 = P(x)$ is a polynomial of degree 3, then $2\frac{d}{dx}\left(y^3.\frac{d^2y}{dx^2}\right)$ is equal to-

(A) $P'''(x) + P'(x)$ (B) $P''(x).P'''(x)$

(C) $P(x).P'''(x)$ (D) A constant

Q44. If $y = \left(x+\sqrt{1+x^2}\right)^n$ then $(1 + x^2)y_2 + xy_1 =$

(A) ny^2 (B) n^2y

(C) n^2y^2 (D) None of these

Q45. If $f(x) = x^n$, then the values of $f(1) - \frac{f'(1)}{1!}+\frac{f''(1)}{2!}-\frac{f'''(1)}{3!}+...+\frac{(-1)^n f^n(1)}{n!}$ is-

(A) 1 (B) 2^n (C) 2^{n-1} (D) 0

Q46. If $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$, where $0 < \theta < \frac{\pi}{2}$, then $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$ is equal to -

(A) $\frac{4\sqrt{2}}{a\pi}$ (B) $\frac{8\sqrt{2}}{a\pi}$

(C) $\frac{4}{a\pi\sqrt{2}}$ (D) None of these

L'Hospital Rule

Q47. If $f(a) = 2$, $f'(a) = 1$, $g(a) = -1$; $g'(a) = 2$, then $\lim\limits_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a} =$

(A) 3 (B) 5 (C) 0 (D) –3

Q48. $\lim\limits_{x\to\pi/6}\left[\frac{3\sin x-\sqrt{3}\cos x}{6x-\pi}\right] =$

(A) $\sqrt{3}$ (B) $1/\sqrt{3}$

(C) $-\sqrt{3}$ (D) $-1/\sqrt{3}$

Q49. $\lim\limits_{n\to\infty}\left[\log_{n-1}(n).\log_n(n+1).\log_{n+1}(n+2)\right.$ $\left.....\log_{n^k-1}\left(n^k\right)\right.$ is equal to

(A) ∞ (B) n

(C) k (D) None of these

Q50. $\lim\limits_{x\to 0^+}\log_{\tan x}(\tan 2x)$ is equal to

(A) 1 (B) –1

(C) 0 (D) None of these

Q51. If $f(1) = 1$, $f'(1) = 2$, then $\lim\limits_{x\to 1}\frac{\sqrt{f(x)}-1}{\sqrt{x}-1} =$

(A) 2 (B) 1

(C) 3 (D) 4

Derivative of Functions Expressed in the Determinant Form

Q52. If f(x), g(x), h(x) are polynomials in x of degree 2 and $F(x) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$ then $F'(x)$ is equal to

(A) 1 (B) 0

(C) –1 (D) None of these

Q53. If $y = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$, then $\frac{dy}{dx}$ is equal to

(A) 1 (B) –1

(C) 0 (D) None of these

Q54. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant . Then $\frac{d^3}{dx^3}(f(x))$ at $x = 0$ is

(A) p (B) $p + p^2$

(C) $p + p^3$ (D) 0

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The slope(s) of common tangent(s) to the curves $y = e^{-x}$ & $y = e^{-x}\sin x$ can be

(A) $-e^{-\pi/2}$ (B) $-e^{-\pi}$ (C) $\frac{\pi}{2}$ (D) 1

Q2. If $f(x) = \sqrt{x^2+6x+9}$, then $f'(x)$ is equal to

(A) 1 for $x < -3$ (B) -1 for $x < -3$

(C) 1 for all $x \in R$ (D) 1 for $x > -3$

Q3. If $y = e^{\sqrt{x}} + e^{-\sqrt{x}}$ then $\frac{dy}{dx}$ equals

(A) $\frac{e^{\sqrt{x}} - e^{-\sqrt{x}}}{2\sqrt{x}}$ (B) $\frac{e^{\sqrt{x}} - e^{-\sqrt{x}}}{2x}$

(C) $\frac{1}{2\sqrt{x}}\sqrt{y^2-4}$ (D) $\frac{1}{2\sqrt{x}}\sqrt{y^2+4}$

Q4. If $y = 2^{3^x}$, then y' equals -

(A) $3^x \ell n3 \ell n2$ (B) $y(\log_2 y) \ell n3 \ell n2$

(C) $2^{3^x} . 3^x \ell n6$ (D) $2^{3^x} . 3^x \ell n3 \ell n2$

Q5. If $y = \tan x \tan 2x \tan 3x$ then $\frac{dy}{dx}$ is equal to-

(A) $3\sec^2 3x \tan x \tan 2x + \sec^2 x \tan 2x \tan 3x + 2\sec^2 2x \tan 3x \tan x$

(B) $2y(\operatorname{cosec} 2x + 2\operatorname{cosec} 4x + 3\operatorname{cosec} 6x)$

(C) $3\sec^2 3x - 2\sec^2 2x - \sec^2 x$

(D) $\sec^2 x + 2\sec^2 2x + 3\sec^2 3x$

Q6. The functions $u = e^x \sin x$; $v = e^x \cos x$ satisfy the equation

(A) $v\frac{du}{dx} - u\frac{dv}{dx} = u^2 + v^2$

(B) $\frac{d^2u}{dx^2} = 2v$

(C) $\frac{d^2v}{dx^2} = -2u$

(D) none of these

Q7. Let $y = \sqrt{x+\sqrt{x+\sqrt{x+.....\infty}}}$ then $\frac{dy}{dx}$ -

(A) $\frac{1}{2y-1}$ (B) $\frac{x}{x-2y}$

(C) $\frac{1}{\sqrt{1+4x}}$ (D) $\frac{y}{2x+y}$

Q8. If $y = \sin^{-1}\left(\frac{5x+12\sqrt{1-x^2}}{13}\right)$, then $\frac{dy}{dx}$ can be equal to -

(A) $\frac{1}{\sqrt{1-x^2}}$ (B) $-\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{3}{\sqrt{1-x^2}}$ (D) None of these

Q9. Two functions f & g have first & second derivatives at $x = 0$ & satisfy the relations,

$f(0) = \frac{2}{g(0)}$, $f'(0) = 2g'(0) = 4g(0)$,

$g''(0) = 5$ $f''(0) = 6$ $f(0) = 3$ then -

(A) if $h(x) = \frac{f(x)}{g(x)}$ then $h'(0) = \frac{15}{4}$

(B) if $k(x) = f(x) . g(x) \sin x$ then $k'(0) = 2$

(C) $\lim\limits_{x\to 0} \frac{g'(x)}{f'(x)} = \frac{1}{2}$

(D) none

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer(s). For each question, choose the option corresponding to the correct answer(s).*

Paragraph # 1 (Ques. 10 to 11)

A curve is represented parametrically by the equations $x = e^t \cos t$ and $y = e^t \sin t$ where t is a parameter. Then -

Q10. The relation between the parameter 't' and the angle α between the tangent to the given curve and the x-axis is given by, '*t*' equals -

(A) $\frac{\pi}{2} - \alpha$ (B) $\frac{\pi}{4} + \alpha$

(C) $\alpha - \frac{\pi}{4}$ (D) $\frac{\pi}{4} - \alpha$

Q11. The value of $\frac{d^2y}{dx^2}$ at the point where $t = 0$ is -

(A) 1 (B) 2

(C) – 2 (D) 3

Paragraph # 1 (Ques. 12 to 14)

Let $\frac{f(x+y)-f(x)}{2} = \frac{f(y)-1}{2} + xy$, $xy \in R$. f(x) is differentiable and $f'(0) = 1$.

Let g(x) be a derivable function at $x = 0$ and follows the functional rule

$$g\left(\frac{x+y}{k}\right) = \frac{g(x)+g(y)}{k} \quad (k \in R, k \neq 0, 2)$$

Let $g'(0) = \lambda \neq 0$.

Q12. Domain of $\ell n(f(x))$ is

(A) R^+ (B) $R - \{0\}$

(C) R (D) R^-

Q13. Range of $y = \log_{3/4}(f(x))$

(A) $(-\infty, 1]$ (B) $\left[\frac{3}{4}, \infty\right)$

(C) $(-\infty, \infty)$ (D) R

Q14. If the graphs of $y = f(x)$ and $y = g(x)$ intersect in coincident points the λ can take values

(A) 3 (B) 1 (C) –1 (D) 4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15.

Column-I	**Column-II**
(A) $y = \cos^{-1}(4x^3 - 3x)$, then $\frac{dy}{dx}$ is	(P) $\frac{3}{1+x^2}, x \in \left(\frac{1}{\sqrt{3}}, \infty\right)$
(B) $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$, then $\frac{dy}{dx}$ is	(Q) $\frac{3}{\sqrt{1-x^2}}, x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$
(R) $\frac{3}{1+x^2}, x \in \left(-\infty, -\frac{1}{\sqrt{3}}\right)$	(S) $-\frac{3}{\sqrt{1-x^2}}, x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right)$

Q16. **Column-I** contains function defined on R and **Column-II** contains their properties. Match them.

Column-I	**Column-II**
(A) $\lim_{n\to\infty}\left(\frac{1+\tan\frac{\pi}{2n}}{1+\sin\frac{\pi}{3n}}\right)^n$ equal	(P) e
(B) $\lim_{x\to0^+}\frac{1}{(1+\text{cosec } x)^{\frac{1}{\ell n(\sin x)}}}$ equals	(Q) e^2
(C) $\lim_{x\to0}\left(\frac{2}{\pi}\cos^{-1}x\right)^{1/x}$ equals	(R) $e^{-2/\pi}$
	(S) $e^{\pi/6}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. If $y=\frac{x^4+4}{x^2-2x+2}$ then $\frac{dy}{dx}$ at $x=\frac{1}{2}$ is 'p', then find the value of 21p.

Q18. If $x^3-y^3+3xy^2-3x^2y+1=0$, then at (0, 1) $\frac{dy}{dx}$ equals

Q19. Let $f(x)=x+\cfrac{1}{2x+\cfrac{1}{2x+\cfrac{1}{2x+.....\infty}}}$

Compute the value of f(100).f'(100).

Q20. If $f(x)=\begin{vmatrix}(x-a)^4 & (x-a)^3 & 1\\ (x-b)^4 & (x-b)^3 & 1\\ (x-c)^4 & (x-c)^3 & 1\end{vmatrix}$

then $f'(x)=\lambda.\begin{vmatrix}(x-a)^4 & (x-a)^2 & 1\\ (x-b)^4 & (x-b)^2 & 1\\ (x-c)^4 & (x-c)^2 & 1\end{vmatrix}$

Find the value of λ.

Q21. If $\lim_{x\to0}\frac{1-\cos x.\cos 2x.\cos 3x.....\cos nx}{x^2}$ has the value equal to 253, find the value of n (where $n\in N$).

Q22. The function $f:R\to R$ satisfies $f(x^2)\cdot f''(x)=f'(x)\cdot f'(x^2)$ for all real x. Given that $f(1)=1$ and $f'''(1)=8$, compute the value of $f'(1)+f''(1)$.

SUBJECTIVE TYPE

Q23. Find differential coefficient of $y=(x)^{\sin x}$

Q24. Find differential coefficient of $y=\left(1+\frac{1}{x}\right)^x$

Q25. If $y=\sin x\cdot\sin 2x\cdot\sin 3x.....\sin nx$, find y'.

Q26. If $f(x)=(x+1)(x+2)(x+3).....(x+n)$ then f'(0) is -

Q27. (i) Let $f(x)=x^2-4x-3$, $x>2$ and let g be the inverse of f.
Find the value of g' where f(x) = 2.

(ii) Let $f:R\to R$ be defined as
$f(x)=x^3+3x^2+6x-5+4e^{2x}$
and $g(x)=f^{-1}(x)$, then find g'(–1).

Q28. If $y=x\ell n[(ax)^{-1}+a^{-1}]$, prove that $x(x+1)\frac{d^2y}{dx^2}+\frac{dy}{dx}x=y-1$

Q29. Find a polynomial function $f(x)$ such that $f(2x)=f'(x)f''(x)$.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let f(x) be a polynomial function of second degree. If f(1) = f(–1) and a, b, c are in A.P. then f'(a), f'(b) and f'(c) are in.

(A) Arithmetic-Geometric Progression

(B) Arithmetic progression (A.P.)

(C) Geometric progression (G.P.)

(D) Harmonic progression (H.P.)

Q2. If $f(x) = \underset{n\to\infty}{Lt}\left(\cos\frac{x}{2}\cos\frac{x}{4}\cos\frac{x}{8}.....\cos\frac{x}{2^n}\right)$, then $f'(x)$ at $x = \frac{\pi}{2}$ is -

(A) $\frac{4}{\pi^2}$ (B) $-\frac{4}{\pi^2}$

(C) 0 (D) None of these

Q3. If $x^y \cdot y^x = 1$, then $\frac{dy}{dx} =$

(A) $\frac{y(x\ell n\, y - y)}{x(y\ell n\, x - y)}$ (B) $\frac{y(x\ell n\, y - y)}{x(y\ell n\, x + x)}$

(C) $\frac{y(x\ell n\, y + y)}{x(y\ell n\, x - x)}$ (D) $\frac{-y(x\ell n\, y + y)}{x(y\ell n\, x + x)}$

Q4. If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$ then $\frac{dy}{dx} =$

(A) $-\frac{y}{x}$ (B) $\frac{y}{x}$ (C) $-\frac{x}{y}$ (D) $\frac{x}{y}$

Q5. If $x = \sec\theta - \cos\theta$, $y = \sec^{10}\theta - \cos^{10}\theta$ and $(x^2+4)\left(\frac{dy}{dx}\right)^2 = k(y^2+4)$, then k is equal to

(A) $\frac{1}{100}$ (B) 1 (C) 10 (D) 100

Q6. If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2 f(x)$, then $\frac{dy}{dx}$ at $x = -1$ is equal to

(A) 0 (B) $\frac{1}{14}$

(C) $-\frac{1}{14}$ (D) None of these

Q7. If $u = f(x^3)$, $v = g(x^2)$, $f'(x) = \cos x$ and $g'(x) = \sin x$, then $\frac{du}{dv} =$

(A) $\frac{1}{2}x\cos x^3 \cdot \operatorname{cosec} x^2$

(B) $\frac{3}{2}x\cos x^3 \operatorname{cosec} x^2$

(C) $\frac{1}{2}x\sec x^3 \sin x^2$

(D) $\frac{3}{2}x\sec x^3 \operatorname{cosec} x^2$

Q8. Derivative of $\tan^{-1}\frac{2x}{1-x^2}$ w.r.t $\sin^{-1}\frac{2x}{1+x^2}$ is ($x \in (-1, 1)$)

(A) $\frac{1}{1+x^2}$ (B) $\frac{1}{1-x^2}$

(C) 0 (D) 1

Q9. If $\ln(x + y) = 2xy$, then $y'(0) =$

(A) 1 (B) –1 (C) 2 (D) 0

Q10. The expression of $\frac{dy}{dx}$ of the function $y = a^{x^{a^{x^{\cdots\cdots\infty}}}}$ is -

(A) $\frac{y^2}{x(1-y\ell n\, x)}$

(B) $\frac{y^2\ell n\, y}{x(1-y\ell n\, x)}$

(C) $\frac{y^2\ell n\, y}{x(1-y\ell n\, x\ \ell n\, y)}$

(D) $\frac{y^2\ell n\, y}{x(1+y\ell n\, x\ \ell n\, y)}$

Q11. If $y=\sqrt{\sin x+\sqrt{\sin x+\sqrt{\sin x+.....\text{to }\infty}}}$, then the value of $\frac{dy}{dx}=$

(A) $\sqrt{\frac{\sin x}{y+1}}$ (B) $\frac{\sin x}{y+1}$

(C) $\frac{\cos x}{2y+1}$ (D) $\frac{\cos x}{2y-1}$

Q12. If f(x) = x + tan x and f is inverse of g then g′ (x) equal to

(A) $\frac{1}{1+[g(x)-x]^2}$ (B) $\frac{1}{2-[g(x)-x]^2}$

(C) $\frac{1}{2+[g(x)-x]^2}$ (D) None of these

Q13. If $f(x)=\tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$, $0\le x<\frac{\pi}{2}$, then $f'\left(\frac{\pi}{6}\right)$ is

(A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Q14. If f (x) = $\sin^{-1}$(sin x) + $\cos^{-1}$(sin x) and ϕ (x) = f (f(f(x))), then ϕ′(x) is equal to

(A) 1 (B) sin x

(C) 0 (D) None of these

Q15. If $y=\tan^{-1}\left(\frac{x}{\sqrt{1+x^2}-1}\right)$, then $\frac{dy}{dx}$ is equal to (x ≠ 0)

(A) $\frac{1}{2(1+x^2)}$ (B) $-\frac{1}{2(1+x^2)}$

(C) $\frac{1+x^2}{2}$ (D) $\frac{-(1+x^2)}{2}$

Q16. The derivative of $\sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$ is

(x > 0, ≠ 1)

(A) $\frac{2}{1+x^2}$ (B) $\frac{-2}{1+x^2}$

(C) $\frac{1}{1+x^2}$ (D) $\frac{-1}{1+x^2}$

Q17. If $y=\tan^{-1}\frac{1}{x^2+x+1}+\tan^{-1}\frac{1}{x^2+3x+3}+\tan^{-1}\frac{1}{x^2+5x+7}+.....$ to n terms then $\frac{dy}{dx}=$ (x > 0).

(A) $\frac{1}{x^2+n^2}-\frac{1}{x^2+1}$

(B) $\frac{1}{(x+n)^2+1}-\frac{1}{x^2+1}$

(C) $\frac{1}{x^2(n+1)^2}-\frac{1}{x^2+1}$

(D) None of these

Q18. Let f(x) = $3x^2$ + 4x g′(1) + g″(2) and g(x) = $2x^2$ + 3x.f′(2) + f″(3) then incorrect option is/are –

(A) f′(1) = 22 + 12f′(2)

(B) g′(2) = 44 + 12g′(1)

(C) f″(3) + g″(2) = 10

(D) f′(3) = 34 + 2f′(2)

Q19. If $y=e^{ax}\sin bx$, then $\frac{d^2y}{dx^2}-2a\frac{dy}{dx}$ is equal to

(A) –(a^2 + b^2) y

(B) (a^2 + b^2) y

(C) – y

(D) None of these

Q20. Let f be a polynomial. Then the second derivative of $f(e^x)$ is

(A) $f''(e^x)e^x + f'(e^x)$

(B) $f''(e^x)e^{2x} + f''(e^x)e^x$

(C) $f''(e^x)$

(D) $f''(e^x)e^{2x} + f'(e^x)e^x$

Q21. If $\frac{d^2x}{dy^2}\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} = k$, then k is equal to

(A) 0

(B) 1

(C) 2

(D) None of these

Q22. $\frac{d^2x}{dy^2}$ equals

(A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

(B) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$

(C) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(D) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

Q23. Let $g(x) = \ell n\, f(x)$ where f(x) is a twice differentiable positive function on $(0, \infty)$ such that $f(x + 1) = x\, f(x)$.

Then for $N = 1, 2, 3, \ldots$; $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$

(A) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots \frac{1}{(2N-1)^2}\right\}$

(B) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots \frac{1}{(2N-1)^2}\right\}$

(C) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots \frac{1}{(2N+1)^2}\right\}$

(D) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots \frac{1}{(2N+1)^2}\right\}$

Q24. Given that $f'(2) = 6$ and $f'(1) = 4$,

then $\lim_{h\to 0} \frac{f\left(2h + 2 + h^2\right) - f(2)}{f\left(h - h^2 + 1\right) - f(1)} =$

(A) does not exist

(B) is equal to –3/2

(C) is equal to 3/2

(D) is equal to 3

Q25. Let

$$f(x) = \begin{vmatrix} \sin 3x & 1 & 2\left(\cos\left(\frac{3x}{2}\right) + \sin\left(\frac{3x}{2}\right)\right)^2 \\ \cos 3x & -1 & 2\left(\cos^2\left(\frac{3x}{2}\right) - \sin^2\left(\frac{3x}{2}\right)\right) \\ \tan 3x & 4 & 1 + 2\tan 3x \end{vmatrix}$$

then the value of $f'(x)$ at $x = (2n + 1)\pi$, $n \in I$ (the set of integers) is equal to -

(A) $(-1)^n$

(B) 3

(C) $(-1)^{n+1}$

(D) 9

Exercise - 2A

Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If 1 is a twice repeated root of the equation $a x^3 + bx^2 + bx + d = 0$ then -

(A) $a = b = d$ (B) $a + b = 0$

(C) $b + d = 0$ (D) $a = d$

Q2. If $y = \log_2\{\log_2 x\}$, then $\frac{dy}{dx}$ is equal to -

(A) $\frac{\log_2 e}{x \log_e x}$ (B) $\frac{1}{x \log_e x \log_e 2}$

(C) $\frac{1}{\log_e (2x)^x}$ (D) none of these

Q3. If $y = \frac{\sqrt{(1+t^2)} - \sqrt{(1-t^2)}}{\sqrt{(1+t^2)} + \sqrt{(1-t^2)}}$ and $x = \sqrt{(1-t^4)}$, then $\frac{dy}{dx}$ is equal to -

(A) $\frac{-1}{t^2\left\{1+\sqrt{(1-t^4)}\right\}}$ (B) $\frac{\left\{\sqrt{(1-t^4)}-1\right\}}{t^6}$

(C) $\frac{1}{t^2\{1+\sqrt{(1-t^4)}\}}$ (D) $\frac{1-\sqrt{(1-t^4)}}{t^6}$

Q4. If $2^x + 2^y = 2^{x+y}$ then $\frac{dy}{dx}$ has the value equal to -

(A) $-\frac{2^y}{2^x}$ (B) $\frac{1}{1-2^x}$

(C) $1-2^y$ (D) $\frac{2^x(1-2^y)}{2^y(2^x-1)}$

Q5. If $y^2 + b^2 = 2xy$, then $\frac{dy}{dx}$ equals -

(A) $\frac{1}{xy-b^2}$ (B) $\frac{y}{y-x}$

(C) $\frac{xy-b^2}{(y-x)^2}$ (D) $\frac{xy-b^2}{y}$

Q6. If g is inverse of f and $f(x) = x^2 + 3x - 3$ $(x > 0)$ then $g'(1)$ equals

(A) $\frac{1}{2g(1)+3}$ (B) -1

(C) $\frac{1}{5}$ (D) $\frac{-f'(1)}{(f(1))^2}$

Q7. If $f : R \to R$ is a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3), x \in R$, then $f(1) - f(0)$ is equal to

(A) -2 (B) $f(2)$

(C) $f(3)$ (D) $f(2)$

Q8. If

$$f(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\beta) & \sin(\alpha-\beta) \end{vmatrix}$$

then $f(\alpha), f(\beta), f(\gamma)$ are in -

(A) A.P. (B) G.P.

(C) H.P. (D) None of these

Q9. Let $f(x) = (ax + b)\cos x + (cx + d)\sin x$ and $f'(x) = x \cos x$ be an identity in x, then -

(A) $a = 0$ (B) $b = 1$

(C) $c = 1$ (D) $d = 0$

Q10. Let $f(x) = x^n$, n being a non-negative integer, the value of n for which the equality $f'(a + b) = f'(a) + f'(b)$ is valid for all $a, b > 0$ is -

(A) 0 (B) 1

(C) 2 (D) None of these

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques.11 to 13)

If $y = f(x)$ be a differentiable function of x such that whose second, third,....., nth derivatives exist. i.e, nth derivative of y is denoted by

$y_n, \frac{d^n y}{dx^n}, D^n y, y^n, f^n(x)$ is defined as

$$\frac{d^n y}{dx^n} = \lim_{h\to 0} \frac{f^{n-1}(x+h) - f^{n-1}(x)}{h}$$

Q11. If $y = e^{3x+7}$, then the value of $y_n(0)$ is -

(A) 1 (B) 3^n

(C) $3^n \cdot e^7$ (D) $3^n \cdot e^7 \cdot 7!$

Q12. If $y = \frac{\ln x}{x}$, then the value of $y''(e)$ is -

(A) 1 (B) $-\frac{1}{e}$ (C) $-\frac{1}{e^2}$ (D) $-\frac{1}{e^3}$

Q13. If $x = \sin t$, $y = \sin kt$, then the value of $(1 - x^2)y_2 - xy_1$ is

(A) k^2y (B) $-k^2y$ (C) ky^2 (D) $-ky^2$

Paragraph # 2 (Ques.14 to 16)

Limits that lead to the indeterminate forms 1^∞, 0^0, ∞^0 can sometimes be solved taking logarithm first and then using L'Hõpital 's rule.

Let $\lim_{x\to a}(f(x))^{g(x)}$ is in the form of ∞^0, it can be written as $e^{\lim_{x\to a} g(x)\ell nf(x)} = e^L$.

Where $L = \lim_{x\to a} \frac{\ell nf(x)}{1/g(x)}$ is $\frac{\infty}{\infty}$ form and can be solved using L'Hõpital 's rule.

Q14. $\lim_{x\to 1^+} x^{1/(1-x)}$ -

(A) –1 (B) e^{-1} (C) –2 (D) e^{-2}

Q15. $\lim_{x\to\infty}\left[(\ell nx)^{1/2x} + x^{1/x^n}\right] \forall\ n \in N$ -

(A) 2 (B) 0 (C) $e^{1/2}$ (D) e

Q16. $\lim_{x\to 0^+}(\sin x)^{2\sin x}$ -

(A) 1 (B) 0

(C) 2 (D) does not exist

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17.

Column-I	Column-II
(A) If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then $\frac{dy}{dx}$ is	(P) $\frac{2}{1+x^2}, \lvert x\rvert < 1$
(B) If $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then $\frac{dy}{dx}$ is	(Q) $\frac{2}{1+x^2}, \lvert x\rvert \neq 1$
(C) If $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then $\frac{dy}{dx}$ is	(R) $-\frac{2}{1+x^2}, x < 0$
	(S) $-\frac{2}{1+x^2}, \lvert x\rvert > 1$
	(T) Non-existent, $\lvert x\rvert = 1$

Q18.

Column-I	Column-II
(A) If $f(x) = x^3 + x + 1$, then $f'(x^2 + 1)$ at $x = 0$ is	(P) 1
(B) If $f(x) = \log_{x^2}(\log x)$, then $f'(e^e)$ is equal to	(Q) 0
(C) For the function $y = \ell n \tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$ if $\frac{dy}{dx} = p\sec x + q$, then $(p + q)$ is equal to	(R) 28
(D) If $f(x) = \lvert x^3 - x^2 + x - 1\rvert \sin x$, then $4f'(28f(f(\pi)))$ is equal to	(S) 4

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. Let f, g and h are differentiable functions. If $f(0) = 1; g(0) = 2; h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are $(f g)'(0) = 6$; $(g h)'(0) = 4$ and $(h f)'(0) = 5$ then compute the value of $(fgh)'(0)$.

Q20. Let P(x) be a polynomial of degree 4 such that $P(1) = P(3) = P(5) = P'(7) = 0$. If the real number $x \neq 1, 3, 5$ is such that $P(x) = 0$ can be expressed as $x = p/q$ where 'p' and 'q' are relatively prime, then $(p + q)$ equals.

Q21. If a curve is represented parametrically by the equations

$$x = \sin\left(t+\frac{7\pi}{12}\right) + \sin\left(t-\frac{\pi}{12}\right) + \sin\left(t+\frac{3\pi}{12}\right),$$

$$y = \cos\left(t+\frac{7\pi}{12}\right) + \cos\left(t-\frac{\pi}{12}\right) + \cos\left(t+\frac{3\pi}{12}\right)$$

then find the value of $\frac{d}{dt}\left(\frac{x}{y}-\frac{y}{x}\right)$ at $t = \frac{\pi}{8}$.

Q22. Let $f(x) = x^2 - 4x - 3,\ x > 2$ and let g be the inverse of f. Find the value of $(888.g')$ at $f(x) = 9$.

SUBJECTIVE TYPE

Q23. Prove that if $|a_1 \sin x + a_2 \sin 2x + ... + a_n \sin nx| \leq |\sin x|$ for $x \in R$, then $|a_1 + 2a_2 + 3a_3 + ... + na_n| \leq 1$

Q24. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3.(x^3 - y^3)$, prove that $\frac{dy}{dx} = \frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$.

Q25. $$f(x) = \begin{cases} b\sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2} < x < 0 \\ 1/2, & \text{at } x = 0 \\ \frac{e^{ax/2}-1}{x}, & 0 < x < \frac{1}{2} \end{cases}$$

If f(x) is differentiable at $x = 0$ and $|c| < 1/2$ then find the value of 'a' and prove that $64b^2 = 4 - c^2$.

Q26. If $x = \frac{1}{z}$ and $y = f(x)$, show that

$$\frac{d^2f}{dx^2} = 2z^3\frac{dy}{dz} + z^4\frac{d^2y}{dz^2}:$$

Q27. If $\cos\frac{x}{2}\cdot\cos\frac{x}{2^2}\cdot\cos\frac{x}{2^3}.....\cos\frac{x}{2^n} = \frac{\sin x}{2^n \sin\left(x/2^n\right)}$, then prove that

$$\sum_{r=1}^{n}\frac{1}{2^r}\tan\frac{x}{2^r} = \frac{1}{2^n}\cot\frac{x}{2^n} - \cot x.$$

Q28. Let f(x) be a derivable function at $x = 0$ & $f\left(\frac{x+y}{k}\right) = \frac{f(x)+f(y)}{k}$ $(k \in R, k \neq 0, 2)$. Show that f(x) is either a zero or an odd linear function.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	A	B	D	B	B	A	A	D	B
11	12	13	14	15	16	17	18	19	20
C	B	A	B	C	C	B	C	B	A
21	22	23	24	25	26	27	28	29	30
B	B	A	B	D	A	C	A	C	B
31	32	33	34	35	36	37	38	39	40
A	C	C	D	A	B	C	B	C	A
41	42	43	44	45	46	47	48	49	50
B	A	C	B	D	B	B	B	C	A
51	52	53	54						
A	B	A	D						

Ex - 1A

1. A,B **2.** B,D **3.** C,A **4.** B,D **5.** A,B,C

6. (A),(B),(C) **7.** A,C,D **8.** (A,B) **9.** A, B, C **10.** C

11. B **12.** C **13.** A **14.** A,C

15. (A) → (Q,S); (B) → (P,R) **16.** (A) → (S); (B) → (P); (C) → (R)

17. 63 **18.** 1 **19.** 100 **20.** 3 **22.** 6

23. $x\sin x\left(\frac{\sin x}{x}+\log x.\cos x\right)$ **24.** $\left(1+\frac{1}{x}\right)^x\left(\log\left(1+\frac{1}{x}\right)-\frac{1}{x+1}\right)$

25. y' = y (cot x + 2 cot 2x + 3 cot 3x + ... + n cot n x)

26. $n!\left(1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n}\right)$ **27.** (i) 1/6, (ii) $\frac{1}{14}$ **29.** $\frac{4x^3}{9}$

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	D	C	D	C	B	D	A	C
11	12	13	14	15	16	17	18	19	20
D	C	D	C	B	B	B	D	A	D
21	22	23	24	25					
A	D	A	D	D					

Ex - 2A

1. B,C,D **2.** A,B **3.** A,C **4.** A,B,C,D **5.** B,C

6. A,C **7.** A,B **8.** A,B,C **9.** A,B,C,D **10.** A,C

11. C **12.** D **13.** B **14.** A **15.** A **16.**A

17. (A) → (P,S,T); (B) → (R); (C) → (P,Q,T)

18. (A) → (S); (B) → (Q); (C) → (P); (D) → (S)

19. 16 **20.** 100 **21.** 8 **22.** 111

-: Important Notes & Formulas :-